

fixed under norm based regime. The tariff responses received from Sasan, Mundra and Krishnapatnam have shown that initial expectations itself were modest and the actual bids received turned out even lower than what was generally expected. The benchmarks achieved in case of UMPPs already bid out heralds the beginning of a new tariff regime, where competition is to be the key driver, efficiency being the only norm and providing reliable power to consumer at affordable and cost effective manner being the only motive. Thus, UMPP initiative has created a new paradigm for power development in the country.

■ PLANNED TRANSMISSION

The Indian power system is demarcated into five independent regional grids - Northern, Eastern, Western, Southern, and North-eastern Regions. Inter-regional connectivity has been planned with hybrid systems consisting of HVDC, Ultra High Voltage AC (765 KV) and Extra High Voltage AC (400 KV) lines. The present inter-regional transfer capacity is 17,000 MW that connects the Northern, Western, Eastern and North Eastern Regions in a synchronous mode. This capacity is further slated to grow to 37,700 MW by 2012. Efforts are underway to have synchronous integration of Southern Region with rest of the Regions, forming an all India synchronous grid.

To promote competition and private sector participation in the transmission segment, Ministry of Power has notified tariff based competitive guidelines for development of transmission projects. According to these guidelines, an Empowered Committee under the chairmanship of Member, Central Electricity Regulatory Commission (CERC) has been constituted. The Committee has identified 14 transmission projects to be developed through tariff based competitive bidding. The Rural Electrification Corporation (REC) and Power Finance Corporation (PFC) have been entrusted with the task of formulating FRs / DPRs for these transmission lines and to invite bids.

■ BRINGING EFFICIENCY IN DISTRIBUTION

Distribution despite being of crucial importance in the entire electricity supply chain, remained neglected area and thus, resulting in huge Aggregate Technical and Commercial (AT&C) losses. The paradox is that to reduce the losses, heavy dose of investment is required to revitalize the distribution segment; at the same time investors don't find it an attractive venture due to prevailing state of huge losses. Furthermore, inadequate investment meant that nothing was spent on improvement works. This resulted in unplanned extensions of the distribution lines, overloading of the system elements like transformers and conductors. In addition to that, low metering efficiency, theft and pilferages, poor energy accounting and billing added further to the commercial losses.

The Ministry of Power realised that reduction of these losses was essential to bring economic viability to the State Utilities. Thus, the Accelerated Power Development and Reform Programme was launched in 2001. The main objective of the programme was to strengthen the Sub-transmission and Distribution network and to bring down the AT&C losses significantly.

The programme has led to reduction in the overall AT&C loss from 38.86 per cent in 2001-02 to 34.54 per cent in 2005-06. The commercial loss of the State Power Utilities reduced significantly during this period from Rs 29,331 crore to Rs 19,546 crore. The loss as percentage of turnover was reduced from 33 per cent in 2000-01 to 16.60 per cent in 2005-06. Although at national level the AT&C losses of State power utilities reduced only marginally, however, the loss has come down significantly in towns where APDRP scheme has been implemented. AT&C losses have been brought down to below 20% in 215 APDRP towns in the country, of which 163 towns have been brought down losses to below 15%.

Drastic reduction of losses is the single most essential measure for turning around the power sector and for attracting much needed investments. It appears that mere infusion of investment through APDRP and formal adherence to reform conditionalities are not sufficient condition for drastic loss reduction. A clear commitment to reduce AT&C losses significantly by the end of 11th Plan from the State Governments, good corporate governance reliable and objectively demonstrable systems of accountability coupled with incentive schemes for staffs are some of the other drivers for improvement. Ministry of Power proposes to restructure the APDRP on these lines.

■ REACHING TO RURAL INDIA

Electricity has never been a pure public good as the principle of exclusion was

invariably applied to the consumers by way of prevailing tariffs. However, by not reaching these large numbers of villages we have literally excluded the vast population of the country from making a rational choice as a consumer. This order needed immediate correction. The National Common Minimum Programme has rightly identified this as a major problem of the country and envisaged for providing electricity to all the rural households within the span of five years. This is a daunting task but achievable.

There is a strong co-relation between penetration of electricity in rural areas and improvement in quality of life and literacy. Human development index is invariably linked to providence of commercial energy especially electricity. Rajiv Gandhi Grameen Vidyutikaran Yojana was launched by Ministry of Power in April 2005 to change the energy landscape of rural India by electrifying all the villages in a time bound manner. RGGVY is part of the Bharat Nirman Programme of the Government of India, which targets to electrify 1.25,000 electrified villages and to give electricity connections to 23.4 million households by 2009. The scheme entails provision of 90% capital subsidy for overall cost of the projects identified under the scheme. Furthermore, electrification of un-electrified below poverty line (BPL households) is being finance through 100% capital subsidy by the Government of India (GoI).

GoI had approved capital subsidy of Rs. 5,000 crore for last 2 years of the 10th Plan of the scheme. Continuation of scheme in 11th Plan was sanctioned on 3rd January, 2008, with the provision of capital subsidy of Rs. 28,000 crores. The strategy of infusion of huge investment by Central Government in the scheme has started showing results already. Under this initiative so far, 45,906 un-electrified villages have been electrified and further 30,275 villages have been intensively electrified. Furthermore, electricity connections have been provided to 24.66 lakh rural households including free connections to 20.28 lakh BPL households. To ensure revenue sustainability franchisees system is being promoted and in 73,422 villages in 14 States franchisees are already in place.

■ HARNESSING RENEWABLE ENERGY

Given the limited amount of domestic conventional energy sources, renewable energy sources gain significance in the Indian context. Out of the total installed capacity of over 1,41,500 MW (excluding captive capacity), 35,379 MW is hydro and about 10,855 MW is contributed by other renewable energy sources, like wind. During the course of the Eleventh Plan, capacity addition through hydro segment has been proposed over 16,000 MW. Capacity addition from other renewable sources, including wind, has been proposed to be about 14,000 MW.

■ ENERGY SAVED IS ENERGY CREATED

The Energy Conservation Act-2001 is already in place to facilitate energy conservation drive in the country. The Bureau of Energy Efficiency (BEE), a nodal statutory body at the central level, has been facilitating in the implementation of the EC Act since 2002. The Integrated Energy Policy (IEP) lays emphasis on energy conservation and efficiency. IEP estimates that 15 per cent saving of energy is possible through DSM. Even Prime Minister Dr. Manmohan Singh stressed on DSM interventions in the Conference of Chief Ministers held on May 28, 2007.

For an efficient electricity usage, BEE has envisaged the following major initiatives:

- **Bachat Lamp Yojana** - to promote energy efficiency and high quality compact fluorescent lamps as replacement for incandescent bulbs in households.
- **Agricultural and Municipal DSM** - to target replacement of inefficient pumpsets, street lighting, etc.
- **Standards and Labeling Scheme** - to lay down minimum energy performance standards for high energy end use equipments and appliances. Star Rating given for refrigerators, air conditioners, fluorescent tube and distribution transformers.
- **Energy Conservation Building Code** - to set minimum energy performance standards for new commercial buildings.

Conclusion: The strategy formed to make power available to all by 2012 focuses on accelerated capacity addition, providing complete access to electricity to the rural India, strengthening the national grid, focused programme to reduce AT&C losses significantly and above all promote efficient use of energy along with its conservation. With premeditated planning on generation and energy conservation, the Union Ministry of Power will undoubtedly reach its grand target in years to come.